Dictionary &
Dictionary Method’s IN Python

What is Dictionary in Python ?

- A dictionary is a built-in data type in Python.
- Dictionaries are used to store data values in key:value pairs.
- In dictionary, Each key acts like a index for its value.
- Keys are used to access values.
- Dictionary items are ordered, changeable, and do not allow duplicates.
- In dictionary, Keys must be Unique but values can be duplicate.

Example :-

thisdict = {“brand” : “Ford”, “model” : “Mustang”, “year” : 1964}

print(thisdict)

print(type(thisdict))

Output :-

{‘brand’ : ‘Ford’, ‘model’ : ‘Mustang’, ‘year’ : 1964}

< class ‘dict’ >

Access Dictionary items :-

You can access the items of a dictionary by referring to its key name, inside square brackets.

Example :-

thisdict = {“brand” : “Ford”, “model” : “Mustang”, “year” : 1964}

print(thisdict[“brand”])

Output :-

Ford

Nested Dictionaries :-

A nested dictionary in Python is a dictionary can contains other dictionaries.

Example :-

```
Students = {
    “student1” : {
        “name” : “Ravi”,
        “age” : 21,
        “branch” : “AIML”
    },
    “student2” : {
        “name” : “sita”,
        “age” : 22,
        “branch” : “CSE”
    }
}
print(Students[“student1”][“name”])
```

Output :-

```
Ravi
```

Dictionary Methods :-

- get()
- items()
- keys()
- values()
- pop()
- popitem()
- setdefault()
- update()
- copy()
- fromkeys()
- clear()

1. **get() :-**
   Returns the value of the specified key.
   Example :-
   car = {“brand” : “Ford”, “model” : “Mustang”}
   print(car.get(“model”))
   Output :-
   Mustang

2. **items() :-**
   Returns key-value pairs as tuples.
   Example :-
   car = {“brand” : “Ford”, “model” : “Mustang”}
   print(car.items())
   Output :-
   dict_items([(‘brand’, ‘Ford’), (‘model’, ‘Mustang’)])

3. **keys() :-**
   Returns all keys in the dictionary.
   Example :-
   data = {“a” : 10, “b” : 20, “c” : 30}
   print(data.keys())
   Output :-
   dict_keys([‘a’, ‘b’, ‘c’])

4. **values() :-**
   Returns all values in the dictionary.
   Example :-
   data = {“a” : 10, “b” : 20, “c” : 30}
   print(data.values())
   Output :-
   dict_values([10, 20, 30])

5. **pop() :-**
   Removes a specific key and returns its value.
   Example :-
   data = {“a” : 10, “b” : 20, “c” : 30}
   data.pop(“a”)
   print(data)

Output :-

{'b' : 20, 'c' : 30}

6. **popitem() :-**

   Removes and returns the last inserted key-value pair.

   Example :-

   ```
   data = {"a" : 10, "b" : 20, "c" : 30}
   data.popitem()
   print(data)
   ```

   Output :-

   {' a' : 10, 'b' : 20}

7. **setdefault() :-**

   Gets a value if key exists, else adds the key with default value.

   Example :-

   ```
   x = {"fruit":"apple","car":"Maruti","mobile":"Oneplus 11R"}
   x.setdefault("bike","bajaj")
   print(x)
   ```

   Output :-

   {'fruit': 'apple', 'car': 'Maruti', 'mobile': 'Oneplus 11R', 'bike': 'bajaj'}

8. **update() :-**

   Used to add or modify multiple key-value pairs.

   Example :-

   ```
   colors = {"color1":"white","color2":"black","color3":"blue"}
   colors.update({"color4":"yellow"})
   colors
   ```

   Output :-

   {'color1': 'white', 'color2': 'black', 'color3': 'blue', 'color4': 'yellow'}

9. **copy() :-**

   Create a copy of the dictionary.

   Example :-

   ```
   original = {"x" : 100, "y" : 200}
   d1 = original.copy()
   ```

```
print(d1)
```

Output :-

```
{'x' : 100, 'y' : 200}
```

10.fromkeys() :-

Returns a dictionary with the specified keys and value.

Example :-

```
roll_numbers = (101,102,103)
roll = dict.fromkeys(roll_numbers)
roll
```

Output :-

```
{101: None, 102: None, 103: None}
```

11. clear() :-

Removes all the elements from the dictionary.

Example :-

```
details = {"a" : 10, "b" : 20}
details.clear()
print(details)
```

Output :-

```
{}
```